COOLSHADE · TRACEABILITY MATRIX

CoolShade traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/03b-traceability.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-30	Amish Chadha	Generated by .kit/trace.py

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Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-004	PR-001	Shaded waiting area	CSH-CAL-001, CSH-BLD-001	Plan layout; sun path check per site	Met: shadow 6.30 m² or more at any sun elevation of 45° or more; room for 8 standing, 3 seated and a wheelchair [A2, A3]. Where the shadow falls depends on the site
UR-004	PR-002	Headroom	CSH-CAL-001, CSH-BLD-001	Model check	Met: 2,223 mm under the flattened ends of the rear knee braces; nozzle tips 2,559 mm or higher [B2, B3]
UR-001	PR-003	Radiant heat reduction	CSH-CAL-001, CSH-STM-001	Literature comparison; later field measurement with a globe thermometer (HeatMap Node method)	Met on published evidence: about 17.3 °C for shade sails in Tempe (Middel et al., 2021); unverified for this fabric
UR-002	PR-004	Air cooling from misting	CSH-CAL-001	Energy balance at TRL 3; later field test	Met: 2.71 °C while spraying at 1 m/s, held to 1.35 m/s; 1.35 °C averaged over the cycle [E2, E3]
UR-003	PR-005	No wetting	CSH-CAL-001, CSH-STM-001	Droplet size data from nozzle maker; later field test	Not verifiable at TRL 3: met only if the nozzle's Dv0.9 is below about 80 µm at 7 bar; no droplet data yet [D3]
UR-005, UR-003	PR-006	Smart activation	CSH-CAL-001, CSH-BLD-001, BOM	Control logic review; later bench test	Met by design: stops within 30 s [I1]; thresholds decided (CSH-DDR-001, D2)
UR-015	PR-007	Water use	CSH-CAL-001, CSH-BLD-001, CSH-STM-001	Nozzle flow calculation	Met: 85 L used, 95 L drawn with the drain-and-refill routine [C2, C5]; holds up to 4.75 L/h per nozzle [C3]
UR-006	PR-008	Energy autonomy	CSH-CAL-001, CSH-BLD-001	Energy budget	Met: 203.8 Wh used against 337.5 Wh harvested; 25 Ah battery (CSH-DDR-002; was 20 Ah, 1.005 days) covers 1.256 design days; with the white enclosure (CSH-DEC-001 D9) no charging in air above about 42.7 °C [F2, F5]
UR-007	PR-009	Water hygiene	CSH-CAL-001, CSH-BLD-001	Design review; later microbiological sampling with a partner lab	At risk:* *line drains in about 30 s with the vacuum breaker and water stays under 24 h old with the routine [C5, C7], but tank water sits at about*

User requirement	Product requirement	Title	Evidence	Verification	Status
					34 °C, in the 20 to 50 °C Legionella* growth range (WHO) [I3]
UR-008	PR-010	Wind resistance	CSH-CAL-001, CSH-BLD-001	Structural calculation at TRL 3	Met on paper: fabric off at 30 m/s, posts 0.14, panel bearer 0.02 and purlins 0.07 utilization [G13]; fabric on at 15 m/s, long beams 0.88 [G14]; anchors need about 2.8 kN each, to be confirmed per site. The original case gives 3.52 on the beams [G7]
UR-010	PR-011	Privacy	CSH-CAL-001, CSH-BLD-001	Design review	Met by design; no status link fitted (CSH-DDR-001, D7)
UR-009, UR-006	PR-012	Electrical safety	CSH-CAL-001, CSH-BLD-001	Design review	Met by design: 15 A terminal fuse, margin 2.0 on the largest current [I2]
UR-014	PR-013	Cost	CSH-CAL-001, CSH-STM-001	BOM	Over the value-engineering target by $107:** estimated $882, 13.8 % over, after the parts added to make the design buildable (CSH-DDR-003); savings worth trying are in CSH-DEC-001 [J1, J2]
UR-012	PR-014	Buildability	CSH-CAL-001, CSH-BLD-001	Assembly review	Met by design: 79 bolts and anchors, 36 of them into rivet nuts, no welds [H2]; the hand rivet nut tool and bench vice were added to the tool list by Amish on 2026-10-02 (CSH-DEC-001); build time unverified
UR-011, UR-015	PR-015	Refill interval	CSH-CAL-001	Water balance	Met: 1.41 design days per full tank [C4]
UR-011	PR-016	Maintenance	CSH-CAL-001, CSH-BLD-001	Design review; later field log	At risk** in hard-water areas: scale can block 0.4 mm orifices; not verifiable at TRL 3
UR-013	PR-017	Movability	CSH-CAL-001, CSH-BLD-001	Mass estimate	Met: heaviest part a front post, 21.0 kg [H1]

Gaps

None.